

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-1973

Parkin 138 kV
April 28, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1973. The Interconnection Customer proposes a 127.4 MW solar plus storage facility interconnecting at the Parkin 138 kV substation in the Virginia area. The request was received on January 02, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 619 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$19,321,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MONEL - LA JOIE 138 KV L	FRANKLIN-ELK GROVE - NEW LONDON 138 KV	156.6	212	7.7
BALCONY FALLS - PLAQUEMINE COGENERATION PLANT 138 KV	S TEXA - NACOOCHEE SUBSTATION 138 KV line	130.3	224	7.2
AIBONITO PDS - INTEL 138 KV L	AIBONITO PDS - SOUTH CREEK 138 KV line	126.2	374	8.2
BUTTE - BALCONY FALLS 138 KV	SOUTH TEXA - NACOOCHEE 138 KV	119.6	234	7.2
PLAQUEMINE COGENERATION PLANT - WAY DAM 138 KV Line	S TEXA - NACOOCHEE 138 KV	115.4	264	7.2
ABONITO PDS - ALDERWOOD 138 KV	AIBONITO PDS - SHENANDOAH 138 KV	114.8	222	5.1
WHITE PINE - CUSHAW 138 KV	S TEXA - NACOOCHEE SUBSTATION 138 KV Line	114.7	274	7.2
WAY DAM - CUSHAW 138 KV	SOUTH TEXA - NACOOCHEE 138 KV L	114.4	276	7.2
DEVIL'S HILL - WHITE PINE 138 KV	NACOOCHEE - SOUTH TEXA 138 KV line	114.2	280	7.2
ISLA GRANDE GIS 11191120 - SNOWDEN 138 KV	SOUTH TEXA - NACOOCHEE 138 KV	113.9	285	7.2
DEVIL'S HILL - ISLA GRANDE GIS 11191120 138 KV line	S TEXA - NACOOCHEE 138 KV line	113.8	286	7.2
BUTTE - WALERGA-GALBRATH 138 KV	S TEXA - NACOOCHEE 138 KV line	109.3	245	7.2
DESERT OUTPOST - MONEL SUBSTATION 138 KV Line	NEW LONDON - FRANKLIN-ELK GROVE 138 KV Line	107.6	655	14.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
RED HILL - TIMKEN MERCY 138 KV	RED HILL - FLUVANNA P.S 345 KV line	106.5	352	5.1
OAK CREEK - BEAR 138 KV line	BEAR - COVER 345 KV	105.1	366	5.3
NACOOCHEE - SOUTH TEXA 138 KV	PARKIN - OPOSSUM CREEK 138 KV	104.5	967	51.7
BANNING - FRANKLIN-ELK GROVE 138 KV	FRANKLIN-ELK GROVE - SOUTH TEXA 138 KV	103.5	509	11.1
BANNING - DESERT OUTPOST 138 KV	FRANKLIN-ELK GROVE - S TEXA 138 KV	103.2	523	11.1
SOUTH CRK - WYTHE 138 KV L	S TEXA - NACOOCEE 138 KV	102.4	629	30.1
GRASSMERE - RED HILL 138 KV	BEAR - FLUVANNA P.S 345 KV	101.1	472	6.6
SHENANDOAH - WALWORTH 138 KV	BEAR - CLOVER 345 KV	100.9	520	7.5
AIBONITO PDS - SHENANDOAH 138 KV L	BEAR - CLOVER 345 KV	100.9	544	7.5
S COLTON - YAPHANK 138 KV	DESERT OUTPOST SUBSTATION - S TEXA 138 KV line	100.4	517	6.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LA JOIE SUBSTATION - MONEL 138 KV	Rebuild line with higher-capacity conductor	\$19,987,000	5% (\$1,071,000)
PLAQUEMINE COGENERATION PLANT - BALCONY FALLS 138 KV	Rebuild line with higher-capacity conductor	\$6,885,000	10% (\$685,000)
AIBONITO PDS - INTEL 138 KV L	Rebuild line with higher-capacity conductor	\$10,787,000	12% (\$1,312,000)
BUTTE - BALCONY FALLS 138 KV	Rebuild line with higher-capacity conductor	\$13,500,000	10% (\$1,344,000)
PLAQUEMINE COGENERATION PLANT - WAY DAM 138 KV	Rebuild line with higher-capacity conductor	\$2,907,000	10% (\$289,000)
AIBONITO PDS - ALDERWOOD 138 KV	Rebuild line with higher-capacity conductor	\$43,750,000	21% (\$9,323,000)
WHITE PINE - CUSHAW 138 KV	Rebuild line with higher-capacity conductor	\$5,047,000	10% (\$502,000)
WAY DAM - CUSHAW 138 KV	Rebuild line with higher-capacity conductor	\$2,096,000	10% (\$209,000)
DEVIL'S HILL - WHITE PINE 138 KV	Rebuild line with higher-capacity conductor	\$7,308,000	10% (\$728,000)
ISLA GRANDE GIS 11191120 - SNOWDEN 138 KV	Rebuild line with higher-capacity conductor	\$3,361,000	10% (\$335,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DEVIL'S HILL - ISLA GRANDE GIS 11191120 138 KV line	Rebuild line with higher-capacity conductor	\$4,402,000	10% (\$438,000)
BUTTE - WALERGA-GALBRATH 138 KV	Reconductor line	\$4,269,000	10% (\$425,000)
DESERT OUTPOST - MONEL 138 KV	Reconductor line	\$1,640,000	5% (\$87,000)
RED HILL - TIMEN MERCY 138 KV	Reconductor line	\$1,037,000	3% (\$34,000)
BEAR - OAK CREEK 138 KV	Reconductor line	\$604,000	12% (\$73,000)
S TEXA - NACOOCHEE 138 KV	Reconductor line	\$2,324,000	10% (\$236,000)
FRANKLIN-ELK GROVE - BANNING 138 KV L	Reconductor line	\$341,000	3% (\$11,000)
BANNING - DESERT OUTPOST 138 KV	Reconductor line	\$4,128,000	3% (\$139,000)
SOUTH CRK - WYTHE 138 KV L	Reconductor line	\$2,521,000	10% (\$240,000)
GRASSMERE - RED HILL 138 KV	Reconductor line	\$1,096,000	6% (\$66,000)
SHENANDOAH - WALWORTH 138 KV	Reconductor line	\$10,529,000	10% (\$1,005,000)
SHENANDOAH - AIBONITO PDS 138 KV	Reconductor line	\$7,184,000	10% (\$686,000)
YAPHANK - S COLTON 138 KV Line	Reconductor line	\$2,175,000	4% (\$83,000)
	Total allocated to project		\$19,321,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.